

Case Study: Pulse

The problem

Systems crash without warning. An employee opens a heavy program alongside a dozen Chrome tabs, resource usage climbs past what the device can handle, and the system slows down, then locks up entirely — costing time, work, and sometimes data. Businesses have no early signal before it happens; they only find out after the crash.

What I did

I'm building a monitoring tool that tracks a system's RAM first, then CPU and GPU, since RAM is the resource most directly tied to active crashes. Instead of a single alarm threshold, I designed five warning stages — normal, noticeable, problematic, critical, and action-needed — because a system running at 80%+ usage is often completely normal, and a blunt "shut it down" alert would just train people to ignore it. The system only force-reboots past 90%, and everything below that is a graduated warning instead of a false alarm. Long-term, I want it to learn each system's own normal baseline over time rather than relying on one fixed number for every machine — for now, I'm shipping rule-based thresholds first to get something real working.

What came of it (so far)

The backend logging system and the landing page are built. Next up: finishing the About, Services, Contact, and profile pages, then moving into model integration, training, and evaluation for the learning-based version.

Bio:

2nd year CS student building a resource-monitoring tool that warns IT teams before their systems crash from overuse — currently in active development.

Contact / CTA:

Following along as I build this? Join the early access waitlist to be first in line when it's ready.
